

Set Data Structure

```
In [1]: s = {}  
s
```

```
Out[1]: {}
```

```
In [2]: type(s)
```

```
Out[2]: dict
```

```
In [3]: s1 = {1,2,3,3}  
s1
```

```
Out[3]: {1, 2, 3}
```

```
In [4]: s2 = set()  
type(s)
```

```
Out[4]: dict
```

```
In [5]: s1
```

```
Out[5]: {1, 2, 3}
```

```
In [6]: s2 = {100,1,34,9,43}  
s2
```

```
Out[6]: {1, 9, 34, 43, 100}
```

```
In [7]: s3 = {2,3.4, True,1+2j,'welcome'}  
s3
```

```
Out[7]: {(1+2j), 2, 3.4, True, 'welcome'}
```

```
In [8]: print(s1)  
print(s2)  
print(s3)
```

```
{1, 2, 3}
```

```
{1, 34, 100, 9, 43}
```

```
{True, 2, 3.4, (1+2j), 'welcome'}
```

```
In [9]: s1
```

```
Out[9]: {1, 2, 3}
```

```
In [10]: s1.add(4)  
s1
```

```
Out[10]: {1, 2, 3, 4}
```

```
In [11]: s1.add(5,6)  
s1
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[11], line 1
----> 1 s1.add(5,6)
      2 s1

TypeError: set.add() takes exactly one argument (2 given)
```

In [12]: s1

Out[12]: {1, 2, 3, 4}

In [13]: s4 = s1.copy()
s4

Out[13]: {1, 2, 3, 4}

In [14]: s4[0]

```
-----
TypeError                                Traceback (most recent call last)
Cell In[14], line 1
----> 1 s4[0]

TypeError: 'set' object is not subscriptable
```

In [15]: s4[:]

```
-----
TypeError                                Traceback (most recent call last)
Cell In[15], line 1
----> 1 s4[:]

TypeError: 'set' object is not subscriptable
```

In [16]: s1

Out[16]: {1, 2, 3, 4}

In [17]: s4

Out[17]: {1, 2, 3, 4}

In [18]: s3

Out[18]: {(1+2j), 2, 3.4, True, 'welcome'}

In [19]: s3.pop()

Out[19]: True

In [20]: s3.pop(2)

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[20], line 1  
----> 1 s3.pop(2)  
  
TypeError: set.pop() takes no arguments (1 given)
```

In [21]: s3

Out[21]: {(1+2j), 2, 3.4, 'welcome'}

In [22]: s3.remove(100)

```
-----  
KeyError                                Traceback (most recent call last)  
Cell In[22], line 1  
----> 1 s3.remove(100)  
  
KeyError: 100
```

In [23]: s3.discard(100)

In [24]: s3

Out[24]: {(1+2j), 2, 3.4, 'welcome'}

In [25]: s3.discard(2)
s3

Out[25]: {(1+2j), 3.4, 'welcome'}

In [26]: s3.discard(200)
s3

Out[26]: {(1+2j), 3.4, 'welcome'}

In [27]: s2.remove(100)

In [28]: s2

Out[28]: {1, 9, 34, 43}

In [29]: print(s1)
print(s2)
print(s3)
print(s4)

```
{1, 2, 3, 4}  
{1, 34, 9, 43}  
{3.4, (1+2j), 'welcome'}  
{1, 2, 3, 4}
```

Set Operation

```
In [30]: a = {1,2,3,4,5}
        b = {4,5,6,7,8}
        c = {8,9,10}
```

```
In [31]: a.union(b)
```

```
Out[31]: {1, 2, 3, 4, 5, 6, 7, 8}
```

```
In [32]: a | b | c
```

```
Out[32]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

```
In [33]: print(a)
        print(b)
        print(c)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

```
In [35]: a.intersection(b)
```

```
Out[35]: {4, 5}
```

```
In [36]: b.intersection(c)
```

```
Out[36]: {8}
```

```
In [37]: print(a)
        print(b)
        print(c)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

```
In [38]: a & b
```

```
Out[38]: {4, 5}
```

```
In [39]: a & b & c
```

```
Out[39]: set()
```

```
In [40]: print(a)
        print(b)
        print(c)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

```
In [41]: a.difference(b)
```

```
Out[41]: {1, 2, 3}
```

```
In [42]: b.difference(c)
```

Out[42]: {4, 5, 6, 7}

```
In [43]: b.difference(a)
```

Out[43]: {6, 7, 8}

```
In [44]: b - c
```

Out[44]: {4, 5, 6, 7}

```
In [45]: print(a)
         print(b)
         print(c)
```

{1, 2, 3, 4, 5}

{4, 5, 6, 7, 8}

{8, 9, 10}

```
In [46]: a.symmetric_difference(b)
```

Out[46]: {1, 2, 3, 6, 7, 8}

```
In [47]: b ^ c
```

Out[47]: {4, 5, 6, 7, 9, 10}

```
In [48]: s1
```

Out[48]: {1, 2, 3, 4}

```
In [49]: frozenset(s1)
```

Out[49]: frozenset({1, 2, 3, 4})

```
In [50]: s5 = frozenset(s1)
```

```
In [51]: s5
```

Out[51]: frozenset({1, 2, 3, 4})

```
In [52]: type(s5)
```

Out[52]: frozenset

```
In [53]: s5
```

Out[53]: frozenset({1, 2, 3, 4})

```
In [54]: type(s5)
```

Out[54]: frozenset

```
In [55]: s5.add(5)
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[55], line 1
----> 1 s5.add(5)

AttributeError: 'frozenset' object has no attribute 'add'
```

```
In [56]: s5.remove(5)
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[56], line 1
----> 1 s5.remove(5)

AttributeError: 'frozenset' object has no attribute 'remove'
```

```
In [57]: s5pop(5)
```

```
-----
NameError                                    Traceback (most recent call last)
Cell In[57], line 1
----> 1 s5pop(5)

NameError: name 's5pop' is not defined
```

```
In [58]: print(a)
         print(b)
         print(c)
```

```
{1, 2, 3, 4, 5}
{4, 5, 6, 7, 8}
{8, 9, 10}
```

```
In [59]: b.symmetric_difference_update(c)
```

```
In [60]: b
```

```
Out[60]: {4, 5, 6, 7, 9, 10}
```

```
In [61]: a2 = {1,2,3,4,5,6,7,8,9}
         b2 = {3,4,5,6,7,8}
         c2 = {10,20,30,40}
```

```
In [63]: a2.issuperset(b2)
```

```
Out[63]: True
```

```
In [64]: a2.isdisjoint(c2)
```

```
Out[64]: True
```

```
In [65]: a2.isdisjoint(b2)
```

```
Out[65]: False
```

```
In [66]: a2 = {1,2,3,4,5,6,7,8,9}
         b2 = {3,4,5,6,7,8}
         c2 = {10,20,30,40}
```

```
In [67]: b2.issubset(a2)
```

```
Out[67]: True
```

```
In [68]: c2.issubset(b2)
```

```
Out[68]: False
```

```
In [72]: a3 = {1,2,3,4,5,6,7,8,9}  
b3 = {30,40,50,60,70,80}  
c3 = {10,20,30,40}
```

```
In [73]: a3.issuperset(b3)
```

```
Out[73]: False
```

```
In [74]: b3.issubset(a3)
```

```
Out[74]: False
```

```
In [75]: c3.isdisjoint(b3)
```

```
Out[75]: False
```

```
In [76]: b3.isdisjoint(a3)
```

```
Out[76]: True
```

```
In [ ]:
```